

Vinay Patil

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[LinkedIn](#) — [LeetCode](#) — [GitHub](#) — [Portfolio](#)

PROFESSIONAL SUMMARY

Performance-driven Software Developer with hands-on experience building full-stack web applications, intelligent automation scripts, and computer vision systems. Proficient in React.js, Node.js, Python, and MySQL, with a proven track record of designing secure, scalable architectures. Adept at leveraging next-generation AI development workflows (Cursor, Windsurf, Claude Code) to accelerate project lifecycles, optimize code efficiency, and solve complex programmatic challenges.

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), Python, Java, C++, C, PHP
- **Frontend Development:** React.js, HTML5, CSS3, Bootstrap
- **Backend Development:** Node.js, Express.js, RESTful APIs, Structured Automation Scripts
- **Databases:** MySQL (Schema Design, Normalization, Query Optimization)
- **Developer Tools & Version Control:** Git, GitHub, VS Code
- **Next-Gen AI Workflows:** Cursor, Windsurf, Claude Code, LangChain, OpenCV

PROJECTS

PersonaAI – Human Behavior Modeling System

Ongoing

Tech Stack: Python, GPT, LangChain, Vector Databases (ChromaDB/Pinecone), NLP

- Architecting an AI-driven digital twin platform leveraging LangChain to model human personality frameworks and simulate dynamic behavioral patterns.
- Engineering an algorithmic profiling module that maps conversational inputs to psychological frameworks, including MBTI and the Big Five models.
- Implementing vector embeddings and semantic search to establish a long-term memory layer, reducing context retrieval latency across user sessions.
- Designing a real-time, context-aware conversational engine capable of maintaining multi-turn dialogue with state management.

Stranger Meet – Anonymous Video Chat Platform

2023 – 2024

Tech Stack: React.js, Node.js, Express.js, MySQL, WebRTC, Socket.io

- Developed a full-stack, real-time anonymous video streaming platform processing peer-to-peer communication with minimal latency.
- Built a secure institutional email verification gate using regex and custom token routing, restricting access strictly to authorized university domains.
- Engineered an engagement-focused retention system featuring user streaks and dynamic database counters, boosting simulated platform activity.
- Integrated Google OAuth 2.0 with strict domain filters to ensure secure, authenticated user onboarding and session management.

Central Order Management System

2024 – 2025

Tech Stack: PHP, MySQL, REST APIs, Bootstrap

- Designed and deployed a centralized multi-tenant order management system, streamlining concurrent transactional workflows across multiple food vendors.
- Constructed robust backend RESTful APIs for asynchronous order processing, live state updates, and real-time tracking.
- Optimized a relational MySQL schema with primary/foreign key indexing and normalization rules, ensuring efficient storage and data retrieval speeds.
- Formulated a unified analytics dashboard that consolidated multi-vendor operations, reducing manual tracking overhead and improving processing efficiency.

Virtual Mouse Using Computer Vision

2022 – 2023

Tech Stack: Python, OpenCV, MediaPipe, NumPy

- Built an AI-driven perceptual computing system that maps real-time human hand gestures to hardware cursor controls powered by a standard webcam input.
- Programmed mathematical coordinate mapping to smoothly translate detected finger coordinates into pixel-perfect screen pointer movements.
- Implemented algorithmic gesture recognition using custom frame-differencing logic to execute system operations like clicking, double-clicking, and scrolling.

Secondary Automation Trackers

- **Instagram Auto Posting System (2023-24):** Developed a Python automation script harnessing scheduling libraries and spreadsheet APIs to handle automated asset queues, reducing manual content overhead.
- **AI Desktop Assistant (2025-26):** Programmed a Python system assistant utilizing speech-to-text NLP libraries to interpret voice commands, automate local system scripts, and query external web APIs.

EXPERIENCE

Programming Instructor / Lecturer

2025 – 2026

Institution of Technical Education — Pune, India — (8 Months)

- Delivered comprehensive lectures and practical engineering labs on software fundamentals, full-stack web development, and object-oriented concepts.
- Executed structured code reviews and debugging sessions for students, instilling industry-standard best practices for clean development and algorithmic efficiency.
- Designed rigorous computer science curriculum, examination papers, and hands-on practical assignments focused on Python, SQL database schemas, and JavaScript application architecture.
- Mentored aspiring software engineers through complex programmatic bugs, breaking down complex principles into actionable development workflows.

President – Tech Innovation Club (TIC)

2024 – 2025

Suman Ramesh Tulsiani Technical Campus — Pune, India

- Managed and led a engineering organization of student developers to foster advanced programming, system design, and open-source contributions.
- Spearheaded development workshops, coding bootcamps, and live hackathons centered around modern Web Development, RESTful APIs, and version control frameworks.
- Provided peer mentorship to junior developers, establishing foundational competence in C, C++, Python, JavaScript, and object-oriented methodologies.
- Orchestrated knowledge-sharing seminars by coordinating with faculty advisors, senior engineers, and industry guest speakers.

RELEVANT COURSEWORK

- **Core Computer Science:** Data Structures & Algorithms (LeetCode), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Software Engineering Fundamentals
- **Systems & Architecture:** Full-Stack Web Architecture, Relational Schema Normalization, RESTful API Design, System Automation, Computer Vision & Perceptual Computing Basics

EDUCATION

Bachelor of Engineering (B.E.) – Computer Engineering

Expected Graduation: 2026

Suman Ramesh Tulsiani Technical Campus, Kamshet (SPPU)

Note: Entered directly into the second year of engineering based on academic merit in Diploma.

Diploma in Computer Science

Graduated: 2023

DKTE Society's Yashwantrao Chavan Polytechnic, Ichalkaranji

Aggregate Score: 8.1 CGPA — Core Focus: Practical Software Development, Operating Systems, OOP, and Data Structures.